CERTA
Certified, Embedded, Risk-aware Test design for Analog circuits

GUIDE

Writing CERTA documents

How the document template works, shown with the template itself

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Summary

Every document of the project starts from `document.md`: a front matter with title, subtitle, author, date, type and reference, followed by a fixed sequence of sections. The same file reads well on GitHub and exports to a PDF with the CERTA title block, the logo in the header and the page count in the footer. This guide is itself written with the template, so it doubles as an example of every element.

1 Front matter

The fields at the top of the file fill the title block of the first page and the running header of the following ones.

Field	Use	Example
`title`	Title of the document	Writing CERTA documents
`subtitle`	One sentence that says what it is about	How the document template works
`author`	One name or a list of names	Telmo Miguel-Medina
`date`	A date, or `today` to use the export date	`today`
`doctype`	Kind of document, shown above the title	Report, Protocol, Guide, Minutes
`reference`	Work package and task it belongs to, shown in the header	WP0 · T0.8

`doctype` and `reference` are optional. A long title can be shortened in the header with `running-title`.

2 Structure

The sections of the template follow the order in which a reader needs them:

1. **Summary**, unnumbered: the answer in three to five lines.
2. **Purpose and scope**: the question and what is left out.
3. **Method**: how the work was done, pointing to files instead of copying them.
4. **Results**: tables and figures, each introduced by a sentence.
5. **Conclusions and next steps**: what follows and which task does it.
6. **References**, unnumbered.

Sections that do not apply are removed rather than left empty. Headings are written without numbers; the export numbers them.

2.1 Elements

Tables, lists, links, code and equations use plain Markdown. An equation is written in LaTeX between dollar signs, for example the risk-weighted coverage of the guide:

$$DC_w(Q) = \frac{\sum_k p_k \, w_k \, P(\text{alarm} \mid f_k, Q)}{\sum_k p_k \, w_k}$$

Code and commands go in fenced blocks:

```
pandoc document.md -d docs/templates/pdf/certa-pdf.yaml -o document.pdf
```

Figures take a caption and a relative path:



Figure 1: Certification icon of the project, from `docs/assets/icons`.

3 Export to PDF

From the repository root, with pandoc and a LaTeX distribution that includes XeLaTeX:

```
pandoc path/to/document.md -d docs/templates/pdf/certa-pdf.yaml -o path/to/document.pdf
```

The defaults file `pdf/certa-pdf.yaml` sets the paper, margins, font and colours, and applies the filter `certa.lua`, which builds the title block from the front matter and loads `certa-header.tex`, the LaTeX preamble that draws the header with the logo and the footer with the page count. The font is Avenir Next; on a system without it, change `mainfont` in the defaults file.

4 Conclusions and next steps

The template gives every document the same look without extra work. The PDF of a document is regenerated whenever its Markdown changes and is committed next to it.

References

- Pandoc User's Guide, section "Defaults files". https://pandoc.org/MANUAL.html#defaults-files